



Research Progress of The Algebraic and Geometric Signal Processing

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Abstract

The investigation of novel signal processing tools is one of the hottest research topics in modern signal processing community. Among them, the algebraic and geometric signal processing methods are shown to be one of the most powerful tools for the representation of the classical signal processing method. In this paper, we provide an overview of recent contributions pertaining to the algebraic and geometric signal processing. Specifically, the paper focuses on the mathematical structures behind the signal processing by emphasizing the algebraic and geometric structure of the signal processing. The two major topics were discussed. First, the classical signal processing concepts are related to the algebraic structures, and the recent results associated with the algebraic signal processing theory are introduced. Second, the recent progress of the geometric signal and information processing representations associated with the geometric structure are discussed. From these discussions, it is concluded that the research on the algebraic and geometric structure of signal processing can help the researchers to understand the signal processing tools deeply, and also help us to find novel signal processing methods in signal processing community. Its practical applications are expected to grow significantly in years to come, given that the algebraic and geometric structure of signal processing offer many advantages over the traditional signal processing.

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1. Introduction

As the core of the signal processing community, the signal processing technology plays an important role in modern society. Almost all of the breakthroughs in signal processing were taken with the emergence of new signal processing tools and technologies [1]. These breakthroughs include various advances and extensions from old techniques to new techniques.

For example, the signal processing techniques have moved from single-rate to multirate processing, from time-invariant to adaptive processing [2], from frequency-domain (the traditional Fourier transform domain) to time-frequency analysis [3] or the fractional Fourier domain analysis [4], and from linear to non-linear signal processing. The recent developments in these areas have not only renovated the theory of signal processing, but also resulted in new tools that find applications in various domains. For instance, the multirate signal processing has triggered the recent advances in modern technology and speech/audio coding, and the fractional Fourier transform in time-frequency analysis has found applications in the radar and medical signal processing areas [5–8].

From the signal processing transform point of view [9], the Fourier transform (FT) has been applied widely in many scientific disciplines, and has played important role in almost all the science and technology domains. However, with the

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extension of research objects and scope, FT has been discovered to have many shortcomings [4]. In order to overcome these shortcomings, a series of novel signal analysis theories and tools have been put forward to process nonstationary or no Gauss signals, such as fractional Fourier transform [4], short-time Fourier transform, time-frequency distribution [3], wavelet transform [2], cyclic statistics, AM/FM signal analysis and linear canonical transform [10] and so on.

On the other hand, from mathematical point of view, a set of signals can be looked as a vector space, and all of the filters can be looked as another space, therefore, the linear signal processing progress can be looked as the actions between the signal space and the filter space. As shown in Ref. [11], the signal models used in signal processing are actually the algebraic objects that have more structures than the vector spaces. And from these mathematical structures we can obtain more insights into the signal processing technology and derive novel signal processing tools [12–17]. It is therefore worthwhile and interesting to investigate the mathematical structures of the signal processing tools and transforms.

The main goal of this paper is to provide an overview of recent developments regarding the algebraic and geometric signal processing technologies and their applications in signal processing community. Section 2 describes the algebraic signal processing (ASP), Section 3 presents the recent results about the geometric signal processing, and the conclusions and potential research directions along this way are provided in Section 4.

2. Algebraic signal processing

The goal of this section is to give a short introduction to the algebraic signal processing (ASP) theory that has been recently developed [11–17]. First a short review of the basic knowledge of algebra is given.

2.1. Algebra

The algebra here refers to the theory of groups, rings, and fields in the modern mathematical field. Their definitions are given as follows [19].

A group is a set G with binary operation, called multiplication, that satisfies

- 1) $a(bc) = (ab)c$, i.e., the associative law.
- 2) A left identity or unit element exists, $e \in G$, i.e., $ea = a$ for $a \in G$.
- 3) For each $a \in G$, a left inverse exists, denoted by a^{-1} , which satisfies $a^{-1}a = e$.

A group G is called Abelian if the group operation is commutative, i.e., if $ab = ba$ holds for all $a, b \in G$.

A ring R is a set with two laws of composition called addition and multiplication that satisfy. For all elements of R ,

- 1) Laws of addition
 - a. Associative law: $a + (b + c) = (a + b) + c$.
 - b. Commutative law: $a + b = b + c$.

c. Solvability of the equation $a + x = b$.

2) Laws of multiplication

a. Associative law: $a(bc) = (ab)c$.

b. Distributive laws: $a(b + c) = ab + ac$, $(b + c)a = ba + ca$.

3) R is called a commutative ring if the commutative law $ab = ba$ holds for all $a, b \in R$.

A field F is a commutative ring with identity in which every nonzero element is invertible.

Let $\mathcal{A}$ be an algebra, a (left) algebra $\mathcal{A}$ module is a vector space $\mathcal{M}$ that permits an operation

$$\mathcal{A} \times \mathcal{M} \rightarrow \mathcal{M}, (a, m) \mapsto am, \quad (1)$$

which satisfies, for $a, b, 1 \in \mathcal{A}$, and $m, n \in \mathcal{M}$,

$$\begin{aligned} a(m + n) &= am + an \\ (a + b)m &= am + bm \\ (ab)m &= a(bm) \\ 1m &= m. \end{aligned}$$

2.2. Algebraic signal processing

2.2.1. Algebraic structure in signal processing

In fact, the signal processing can be regarded as the signal passing through a filter, as shown in the following [11].

$$\text{signal} \longrightarrow \boxed{\text{filter}} \longrightarrow \text{signal} \quad (2)$$

From the mathematical point of view, the set of signals is considered to be a vector space. By applying the laws of the vector spaces, the signals can be added and can be multiplied by a scalar α to yield a new signal. Formally

$$\begin{aligned} \text{signal} + \text{signal} &= \text{signal}, \\ \alpha \cdot \text{signal} &= \text{signal}. \end{aligned} \quad (3)$$

And the set of filters can also be regarded as a space, furthermore, the set of the filters have two operations,

$$\begin{aligned} \text{filter} \cdot (\text{signal} + \text{signal}) \\ = \text{filter} \cdot \text{signal} + \text{filter} \cdot \text{signal}. \end{aligned} \quad (4)$$

And the filter themselves can be combined to form new filters,

$$\begin{aligned} \text{filter} + \text{filter} &= \text{filter}(\text{parallel connection}), \\ \text{filter} \cdot \text{filter} &= \text{filter}(\text{series connection}), \\ \alpha \cdot \text{filter} &= \text{filter}(\text{amplification}). \end{aligned} \quad (5)$$

Mathematically, the above structure is described by regarding the filter space as an algebra $\mathcal{A}$ that operates on the signal vector space $\mathcal{M}$, thus making the signal space $\mathcal{M}$ an $\mathcal{A}$ -module [12].

set of filters/linear systems = an algebra $\mathcal{A}$
set of signals = an $\mathcal{A}$ – module $\mathcal{M}$

From the above discussion, it is clear that the linear signal processing can be represented by the algebraic structure, and the advantages can be summarized as follows [11–14].

It should be mentioned that the algebraic approach of signal processing provides the common underpinning for many different infinite and finite linear signal processing schemes, showing, for example, that the spectral transforms are taken as the instantiations of the same common theory. Further, the connection between algebra/modules and signal processing goes in both directions, namely, in the direct direction specifying algebra, and a module provides the ingredients to develop the extensions to the existing signal processing tools. The algebraic theory makes the derivation of algorithms concise and transparent [17], gives insight into the algorithms' structure, enables the classification of the many existing algorithms, and enables the discovery of new algorithms for existing transforms and new linear transforms.

2.2.2. Signal model

Based on the above facts associated with the signal processing, the centre of the algebraic theory of signal processing [11] is the concept of the signal model. It is defined [11–14] as a triple $(\mathcal{A}, \mathcal{M}, \Phi)$, where $\mathcal{A}$ is the chosen algebra of filters, i.e., a vector space where multiplication of the filters is also defined. $\mathcal{M}$ is an $\mathcal{A}$ -module of signals, i.e., a vector space of which the elements can be multiplied by the elements of $\mathcal{A}$. We say that $\mathcal{A}$ operates on $\mathcal{M}$ through filtering. Φ is the generalization of the z-transform. It is defined as a bijective mapping from a vector space V of signal samples in the module $\mathcal{M}$ of signals. The relationship among the triple is shown in Fig. 1 [18].

2.2.3. Shift operator

When a signal model is determined, the Fourier transform, the filtering and the spectrum will be well defined accordingly [18]. It is shown in Ref. [18] that the generalized time shift operator plays an important role in producing the signal model (See Fig. 1), and different shift operator produces different signal model. It is easy to derive that every signal model $(\mathcal{A}, \mathcal{M}, \Phi)$ with shift invariance property has necessarily a commutative algebra $\mathcal{A}$. Furthermore, if the signal model is an addition finite model, the algebra $\mathcal{A}$ has to be a polynomial algebra [18]. The fact helps us to analyze the signal model with polynomial algebra.

In Refs. [11–18], the signal models were derived from basic shift operator, and the shift plays a fundamental role in many areas, including random processes, statistics and dynamical systems, and information theory. As an abstract concept, once there is a group structure, the shifts can be defined for time or space, or in multiple dimensions. In the

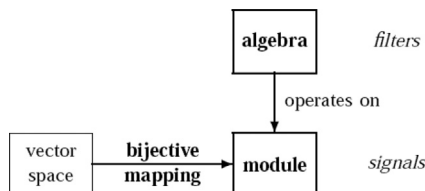


Fig. 1. Relationship among algebra, vector space and module.

algebraic theory, the shift has a particular simple interpretation: it is the generator of the filter algebra. Specifically, we describe a procedure that, starting from the definition of the shift, produces the infinite and finite signal models, and that reveals the degrees of freedom that are available in this construction.

One of the simple shift operators can be defined as [18]

$$q \diamond t_n = t_{n+1}, \quad n \in \mathbb{Z}, \quad (6)$$

where q is the shift operator, $\diamond$ is the shift operation, t_n is the discrete time point. We obtain the classical time models with the associate infinite and finite z-transforms. If a different shift operator is taken as

$$q \diamond t_n = \frac{1}{2}(t_{n-1} + t_{n+1}), \quad (7)$$

then we obtain the well known infinite and finite discrete triangle transforms.

2.3. Special cases of algebraic signal processing

After the definition of the shift operator and the signal model, the classical concepts in signal processing, such as Fourier transform, filtering and spectrum, are well defined. The relationship between the classical signal processing concept and the algebraic signal processing concept is derived in Ref. [11] and shown in Table 1.

2.3.1. 1-D time signal model

A 1-D time signal model proposed in Ref. [11] is introduced here. The signal model can be defined as the infinite 1-D time signal model and finite 1-D signal model based on the discrete signal points.

1) Infinite 1-D time signal model

In model, the shift operator $\diamond$ is defined as

$$q \diamond t_n = t_{n+1}, \quad n \in \mathbb{Z}, \quad (8)$$

where q is the shift operator, t_n is the label of time, $n \in \mathbb{Z}$. Following this definition, the k th order shift operator q_k is defined as

$$q_k \diamond t_n = t_{n+k}. \quad (9)$$

Obviously, the relationship between q and q_k is $q_k = q^k$.

The effect of the shift operator on the set $\sum s_n t_n$ can be defined as

$$q \diamond s = \sum s_n (q \diamond t_n). \quad (10)$$

Let $q = x$, and the initial time $t_0 = 1$, then the time shift operation (Eq. (9)) can be written as

$$t_{n+1} = x \cdot t_n \quad (11)$$

and the solution of Eq. (11) can be derived as $t_n = x^n$.

Table 1

Relationship between signal processing and the algebraic signal processing [11].

Signal processing concept	Algebraic concept (coordinate free)	In coordinates
Filler	$h \in \mathcal{A}$ (Algebra)	$\varphi(h) \in \mathbb{C}^{I \times I}$
Signal	$s = \sum s_i b_i \in \mathcal{M}(\mathcal{A} - \text{module})$	$s = (s_i)_{i \in I} \in \mathbb{C}^I$
Filtering	$h \cdot s$	$\varphi(h) \cdot s$
Impulse	Base vector $b_i \in \mathcal{M}$	$\mathbf{b}_i = (\dots, 0, 1, 0, \dots)^T \in \mathbb{C}^I$
Impulse response of $h \in \mathcal{A}$	$h \cdot b_i \in \mathcal{M}$	$\varphi(h) \cdot \mathbf{b}_i \in \mathbb{C}^I$
Fourier transform	$\Delta: \mathcal{M} \rightarrow \bigoplus_{\omega \in W} \mathcal{M}_\omega$	$\mathcal{F}: \mathbb{C}^I \rightarrow \bigoplus_{\omega \in W} \mathbb{C}_\omega^d \Leftrightarrow \varphi \rightarrow \bigoplus_{\omega \in W} \varphi_\omega$
Spectrum of signal	$\Delta(s) = (s_\omega)_{\omega \in W}$	$\mathcal{F}(s) = (s_\omega)_{\omega \in W}$
Frequency response of $h \in \mathcal{A}$	n.a.	$(\varphi_\omega(h))_{\omega \in W}$
Shift(s)	Generator(s) x of $\mathcal{A}$	$\varphi(x)$
Shift-invariance	$\mathcal{A}$ is commutative	n.a.

Finally, based on Eqs. 8–11, the infinite 1-D time signal model can be derived

$$\mathcal{A} = \left\{ h = \sum_{n \in \mathbb{Z}} h_n x^n \mid \vec{h} = (\dots, h_{-1}, h_0, h_1, \dots) \in l^1(\mathbb{Z}) \right\}, \quad (12)$$

$$\mathcal{M} = \left\{ s = \sum_{n \in \mathbb{Z}} s_n x^n \mid \vec{s} = (\dots, s_{-1}, s_0, s_1, \dots) \in l^2(\mathbb{Z}) \right\}, \quad (13)$$

$$\Phi: V \rightarrow \mathcal{M}, \vec{s} \mapsto \sum s_n x^n. \quad (14)$$

And the corresponding Fourier transform Δ associate with the above signal model can be derived [18]

$$\begin{aligned} \Delta: \mathcal{M} &\rightarrow \left(\bigoplus_{\omega \in W} \mathbb{C} \right) = \mathbb{C}^W, \\ s = S(z) &\mapsto (S(e^{j\omega}))_{\omega \in W} = \omega \mapsto S(e^{j\omega}), \end{aligned} \quad (15)$$

and

$$S(\omega) = \langle s, f_\omega(x) \rangle = \sum_{k \in \mathbb{Z}} s_k e^{-j\omega k}, \quad (16)$$

where $f_\omega(x) = \sum_{k \in \mathbb{Z}} e^{j\omega k} x^k$ is the eigen function of infinite 1-D time signal model [11]. Obviously this model is exactly the classical discrete time Fourier transform.

2) Finite 1-D time signal model

The finite 1-D time signal model can be derived by the similar method with the infinite 1-D time signal model. But the sampling time point $t_0, \dots, t_{n-1}$ is finite. In this case, $x \cdot x^{n-1} = x^n \notin \mathbb{C}_n[x]$, therefore the time shift in $\mathbb{C}_n[x]$ is not a satisfactory operation, and the filtering operation in $\mathbb{C}_n[x]$ cannot be defined.

In order to overcome this problem, the following boundary conditions should be given [18]

$$x^n = r(x) = \sum_{0 \leq k < n} \beta_k x^k, \text{ or } x^n - r(x) = 0. \quad (17)$$

And we can obtain

$$x^{n+k} \equiv r_k(x) \bmod (x^n - r(x)). \quad (18)$$

Form the algebra point of view, the condition Eqs. (17) and (18) let the module changes to

$$\mathcal{M} = \mathbb{C}[x]/(x^n - r(x)). \quad (19)$$

Therefore, the finite 1-D time signal model can be derived

$$\mathcal{A} = \mathcal{M} = \mathbb{C}[x]/(x^n - a), \quad (20)$$

$$\Phi: V \rightarrow \mathcal{M}, \text{ or } \vec{s} \mapsto \sum_{0 \leq k < n} s_k x^k \in \mathcal{M}. \quad (21)$$

The Fourier transform of the above 1-D time signal model can be derived by Chinese Remainder Theorem (CRT). Let $a = 1$, then the Fourier transform Δ based on the signal model $\mathcal{A} = \mathcal{M} = \mathbb{C}[x]/(x^n - 1)$ can be derived:

$$\Delta: \mathbb{C}[x]/(x^n - 1) \rightarrow \bigoplus_{0 \leq k < n} \mathbb{C}[x]/(x - \omega_n^k), \quad (22)$$

$$s = s(x) \mapsto (s(\omega_n^0), \dots, s(\omega_n^{n-1})) \quad (23)$$

It is easy to show that Eqs. (22) and (23) are the classical discrete Fourier transform (DFT) pair.

2.3.2. 1-D space signal model

1-D space signal model [12] can be obtained if the space shift operator is defined as :

$$q \diamond t_n = \frac{1}{2}(t_{n-1} + t_{n+1}), n \in \mathbb{Z}, \quad (24)$$

where t_n is the space label, and q is the suitable space operator.

If the sampling time t_n is infinite, for different type Chebyshev polynomials $C \in \{T, U, V, W\}$, the following infinite 1-D space signal model can be obtained

$$\begin{aligned} \mathcal{A} &= \left\{ h = \sum_{k \geq 0} h_k T_k(x) \mid \vec{h} \in l^1(\mathbb{N}) \right\} \\ \mathcal{M} &= \left\{ s = \sum_{n \geq 0} s_n C_n(x) \mid \vec{s} \in l^2(\mathbb{N}) \right\} \\ \Phi: l^2(\mathbb{N}) &\rightarrow \mathcal{M}, \vec{s} \mapsto \sum_{n \geq 0} s_n C_n(x) \end{aligned} \quad (25)$$

When the sampling time t_n is finite, the different boundary conditions ($p \in \{C_n - C_{n-2}, C_n, C_n \pm C_{n-1}\}$) and the different type Chebyshev polynomials $C \in \{T, U, V, W\}$ are chosen to obtain the following 1-D finite space signal model.

$$\begin{aligned}
\mathcal{A} &= \mathbb{C}[x]/p(x) = \left\{ h = \sum_{0 \leq k < n} h_k T_k(x) \mid h_k \in \mathbb{C} \right\} \\
\mathcal{M} &= \mathbb{C}[x]/p(x) = \left\{ s = \sum_{0 \leq k < n} s_k C_k(x) \mid s_k \in \mathbb{C} \right\} \\
\Phi: \mathbb{C}^n &\rightarrow M, \vec{s} \mapsto \sum_{0 \leq k < n} s_k C_k(x)
\end{aligned} \quad (26)$$

Based on the 1-D space signal model, the second type DCT (Discrete Cosine Transform) and the eighth type DST (Discrete Sine Transform) associated with the algebraic signal processing were investigated in detail in Refs. [12,18], the more detail theory about the algebraic signal processing was described in Refs. [12–20].

2.4. Further research

From above discussion, it can be known that the algebraic signal processing provides an abstract analysis of the commonly signal processing transforms. This representation of signal processing can help us to understand the well-known signal processing transform deeply, and the new or novel signal processing transforms and the fast computation methods of the signal processing transform can be obtained by analyzing the mathematical structure of the signal processing transforms [15–20].

The following trends of algebraic signal processing should be considered in the future research.

- 1) Besides the polynomial algebra, many other types of the algebras also exist, such as group algebra. Therefore, the research on the suitable algebras for the analysis of the algebraic signal processing is one of the future research topics.
- 2) The algebraic signal representation of the well-known signal processing tools, such as wavelet transform, Wigner–Ville transform and ambiguity function, is the other research topic in this field.
- 3) The algebraic representation of the fractional signal processing tools is also one of the most important research topics in algebraic signal processing community, such as fractional Fourier transform, linear canonical transform and fractional calculus.

3. Geometric signal processing

3.1. Information geometry and its applications

Information geometry is a new research field in which the new geometric theory is applied to solve the problems in information and the relative fields. It has been paid much more attention by Japan, Europe, the United States of America and so on. More and more institutes and companies are increasing their supports in researching on Information Geometry in order to make full use of it. Now the information geometry has been applied to statistical inference [21], error correction codes [22], control theory [23–25], statistical physics [26,27], neural networks [28], blind signal separation [29] and so on.

Also, Lie group and the set of the positive definite matrices are used to construct the geometric structures of information geometry and are used for the signal processing, for instance, in Doppler Radar detection [30–33].

Rao [34] considered the set of probability density functions $p(x; \theta)$ with the Fisher Information matrix as a Riemannian manifold $M = \{p(x; \theta)\}$, which is called statistical manifold.

For example, the two-dimensional statistical manifold consists of all normal distributions, which is denoted by

$$M = \left\{ p(x; \mu, \sigma^2) \mid p(x; \mu, \sigma^2) = \frac{-1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\} \right\} \quad (27)$$

where μ is the mean, and σ is the variance. It is easy to show that M is a hyperbolic space with the Gauss curvature $-1/2$. Chentsov [35] and Amari [36] gave the dual structure of a statistical manifold and introduced many algorithms, in which the Kullback–Leibler divergence is

$$K(p, q) = E \left[\log \frac{p}{q} \right] \quad (28)$$

where E denotes the expectation with respect to the probability density function, and p, q are in the statistical manifold, acted as a distance function.

Amari [21] studied a parametric family of systems, which is widely used in systems theory to specify the systems and study their characteristics. Such a parametric family usually forms a finite-dimensional manifold imbedded in the set of all the linear systems. It is useful for investigating the geometric properties of such a parametric family. It is interesting and useful to study whether a natural and invariant distance or divergence measure, a notion of orthogonality, and a notion of straightness or curvature exist in a manifold of systems. Moreover, provided that such geometric structures exist, it is interesting to know if they are useful for the approximation problem, identification problem, robust problem, and many other problems in systems theory. He studied the invariant differential geometric structures inherent in a manifold of systems, and showed that two affine connections $\nabla^{(\alpha)}$, $\nabla^{(-\alpha)}$ satisfy

$$Xg(Y, Z) = g(\nabla^{(\alpha)}_X Y, Z) + g(Y, \nabla^{(-\alpha)}_X Z) \quad (29)$$

where $g = (g_{ij})$ defined by the Fisher information

$$g_{ij}(\theta) = E \left[\frac{\partial}{\partial \theta^i} \log p(x, \theta) \frac{\partial}{\partial \theta^j} \log p(x, \theta) \right] \quad (30)$$

is dual with respect to a Riemannian metric g , and the divergence measures are introduced naturally in such a way that

$$D(p, q) = f(\theta_p) + g(\eta_q) - \theta_p \bullet \eta_q \quad (31)$$

where $f(\theta)$, $g(\eta)$ are dual convex functions. It should be noted that the divergence D is non-negative, but it does not satisfy the triangle inequality and the symmetry, in other words, it is

not the canonical distance function. Also he considered the problems of approximation, identification, and stochastic realization of systems in the framework of information geometry. The approximation problem is explicitly solved by using the notion of α -projection. Some parametric families are α -flat, e.g., an AR model is 1-flat and an MA model is -1 -flat, and the orthogonal decomposition theorem of approximation errors holds in such a family of nesting α -flat models, which show the usefulness of the present geometric framework.

When a parameter space has a certain underlying structure, the ordinary gradient of a function $\nabla f(\theta_t)$ does not represent its steepest direction, but the natural gradient $g^{-1}\nabla f(\theta_t)$ does. In other words, in order to get the minimum of a cost function on a Riemannian space, instead of the ordinary gradient steepest algorithm, the natural gradient steepest algorithm is effective, and the iterative formula is given by

$$\theta_{t+1} = \theta_t - \rho g^{-1}\nabla f(\theta_t) \quad (32)$$

where $0 < \rho < 1$ is the learning rate. The information geometry is used for calculating the natural gradients in the parameter space. The dynamical behavior of natural gradient online learning is analyzed and is proved to be Fisher efficient, implying that it has asymptotically the same performance as the optimal batch estimation of parameters. In Ref. [37], the information geometry was used to study the natural gradient works in Learning.

The classical information geometry is concerned in the cases of random situations. Some non-random cases also can be considered. Especially, matrix Lie group is

$$G(n, C) = \{A \in M^{n \times n} | A| \neq 0\} \quad (33)$$

and the set $PD(n)$ of all positive definite matrices play the important roles for the modern signal processing. The subgroups of general linear group, such as the special orthogonal group

$$SO(n) = \{A \in G(n, R) | A^T A = I, \det(A) = 1\} \quad (34)$$

the special unitary group

$$SU(n) = \{A \in G(n, C) | \bar{A}^T A = I, \det(A) = 1\} \quad (35)$$

and the special symplectic group

$$\begin{aligned} Sp(n) &= \{A \in Sp(n, C) \cap U(2n)\}, Sp(n; C) \\ &= \left\{A \in GL(n; C) | J^T A J = A, \quad J = \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}\right\} \end{aligned}$$

are all compact, and the geodesic curves on them can be extended infinitely so that we can define the geodesic distance, measuring the shortest distance of two points on each group. And such compact Lie groups have non-negative sectional curvatures. In addition, as a submanifold of the general linear group, the set $PD(n)$ of all positive definite matrices is not a subgroup, but a complete submanifold with negative sectional curvature and geodesic curves and geodesic distance can be

defined globally on it. For above compact Lie groups, a bi-invariant Riemannian metric can be defined, which can be used to obtain the natural gradient algorithm in order to get the minimum of a cost function. Similarly, for the set $PD(n)$ of all positive definite matrices, a Riemannian metric can be obtained by using the Frobenius inner product

$$g(X, Y)_A = \text{trace}(A^{-1} X A^{-1} Y) \quad (36)$$

These theories have been applied to the Doppler Radar [30,31], the optimization [38], optimal control [39], compressed sensing [40] and so on.

3.2. Applications of information geometry in signal processing

Neural network has potential application in signal processing. Based on information geometry, Amari and Clichocki ([29]) presented learning algorithms and underlying basic mathematical ideas for adaptive blind signal processing, especially instantaneous blind separation and multichannel blind deconvolution of independent source signals. The information geometry is used to obtain the adaptive learning algorithms based on the natural gradient approach and their properties concerning convergence, stability, and efficiency.

A new on-line learning algorithm which minimizes a statistical dependency among outputs for blind separation of mixed signals was derived by using the Kullback–Leibler divergence [41]. The dependency is measured by the average mutual information (MI) of the outputs. The source signals and the mixing matrix are unknown except for the number of the sources.

A Boltzmann machine is a network of stochastic neurons. The set of all the Boltzmann machines forms a manifold, where the modifiable synaptic weights of connections act as a coordinate system to specify the networks. In Ref. [42], the authors established a natural invariant Riemannian metric and a dual pair of affine connections on The Boltzmann neural network manifold by using information geometry. The meaning of the geometrical structures is elucidated from the stochastic and the statistical point of view. This leads to a natural modification of the Boltzmann machine learning rule.

A main issue of high resolution Doppler radar detection is related to robust statistical estimation of Toeplitz Hermitian positive definite covariance matrices of time series. In Refs. [30,31], the authors considered this problem in the framework of Riemannian symmetric spaces and the framework of information geometry. Based on Frechet-Karcher barycenter definition and geodesics in Bruhat-Tits space, they addressed the problem of N covariance matrices mean estimation. Based on this new definition of N matrices Barycenter and robust distance between 2 matrices, they defined a “Matrix CFAR” that improves the classical approach based on Doppler Filter bank or FFT. They obtained the results of real Doppler radar data in X band. For radar, ones have to process the partial complex data for array processing or the time complex data for Doppler processing: the covariance matrices of these complex

data are Toeplitz Hermitian positive definite matrices. The new approach is pure geometric and uses specific geometry of Hermitian positive definite matrices space to improve the classical approaches.

In Refs. [32–43], the authors studied the target detection of the pulsed-Doppler radar and gave some simulations by using the information geometry.

3.3. Further research

Information geometry deals with the curved spaces with random or non-random cases. Especially, in the curved space, by introducing the Fisher information metric on the statistical manifolds or the left invariant metric, we can construct the iterative formulas for calculating the minimum of a cost function by means of the natural gradient. The following researches are interesting and expected:

- 1) Signal processing by information geometry. For the fractional Fourier transformation, we can define the metric by using the power spectral density function and do the further research. By defining the inner product on matrix Lie groups, we can get a Riemannian metric which can be used to formulate the geometric structure for the research of the blind source separation. Also, we can study Doppler signal and the compressed sensing by using the information geometry since the matrix group is included in the theory.
- 2) So far, the geometric frame of information geometry is constructed by using the most fundamental concepts; therefore, the deep parts of differential geometry, Lie group, fiber bundle, homology groups and so on should be included. This is a valuable work to construct the wider theory framework of information geometry.

4. Conclusions

In this paper, we introduce the algebraic and the geometric approaches of signal processing. In Section 2, the canonical methods of signal processing are recalled, and then the main concepts of the algebraic methods of signal processing, including some important applications, are introduced. In Section 3, the method of information geometrical signal processing is introduced. We recall the concepts of information geometry, including the classical information geometry and the information geometry based on matrix group. Also we introduce the applications of information geometry in several fields, including signal processing, blind signal separation, neural network, Boltzmann machine, Doppler radar detection, fractional Fourier transformation and so on. At last, we propose some problems for the further consideration.

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